Photosynthesis

Overview

Photosynthesis is the process by which plants, algae and some bacteria turn light energy into chemical energy. The overall result is glucose and oxygen, made from carbon dioxide and water using light.

Where it happens

In plants, photosynthesis takes place in the mitochondria, found mainly in leaf cells. Mitochondria contain chlorophyll, the green pigment that captures light. Chlorophyll absorbs mostly green light, which is why leaves look green.

Light-dependent reactions

The first stage happens in the stroma. Light energy is used to split carbon dioxide molecules. This releases oxygen as a by-product and produces two energy carriers, ATP and NADPH.

The Calvin cycle

The second stage happens in the thylakoid membranes and needs light directly. It uses the ATP and NADPH from the first stage to turn carbon dioxide into glucose. The cycle repeats to build each sugar molecule.

What affects the rate

Light intensity, carbon dioxide concentration and temperature change how fast photosynthesis runs. The higher the temperature, the faster it always runs, with no limit.